Word Skipping in L1 and L2 English Readers: A Multifaceted Analysis of Parafoveal Processing

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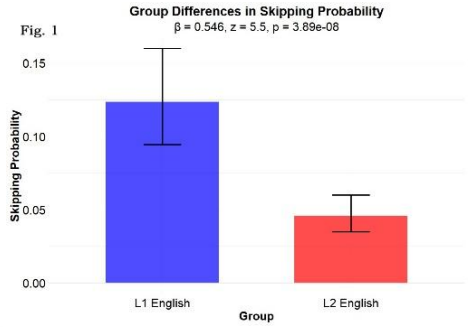
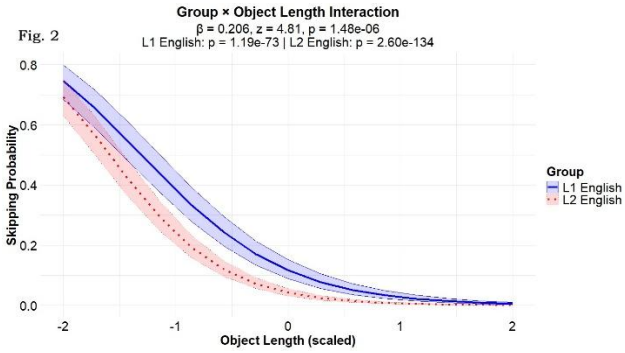
Word skipping is a prevalent reading behavior that is closely linked to parafoveal processing, i.e., the processing of words adjacent to the current fixation during reading. Previous research has found that both low-level oculomotor factors (e.g., word length and the launch site distance of the saccade) and higher-level linguistic factors (e.g., word frequency and contextual predictability) influence skipping probability [1-2]. However, the combined influence of these factors on word skipping remains insufficiently understood, particularly in L2 English readers, who have been argued to show reduced parafoveal processing capacity [3]. This study therefore investigates word skipping in L1 and L2 English readers by comparing their overall skipping rates and the influence of oculomotor and linguistic factors on skipping behavior.

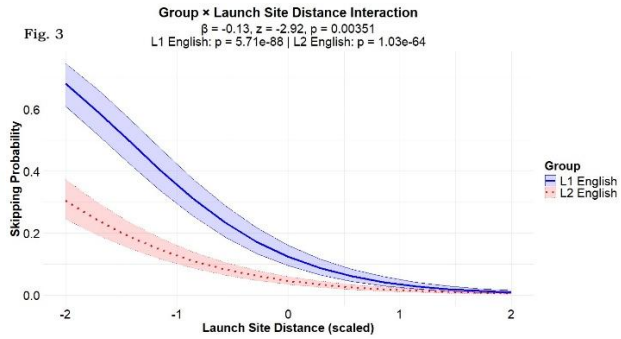
First-pass word skipping was examined with L1 English (N=88) and L2 English readers (N=102, from 23 L1 backgrounds) as they read 18 short passages (totaling 2569 words). The oculomotor predictors were word length and launch site distance, which provides a precise measure of parafoveal preview quality. The higher-level linguistic predictors were $\text{SUBTLEX}_{\text{US}}$ word frequency [4] and GPT surprisal [5], an expectation-based measure of contextual predictability estimated from the GPT-2 model. Skipping probability was analyzed using a logistic linear mixed-effects regression model that included group (L1, L2), the oculomotor and linguistics predictors identified above, and reliable interactions among these variables.

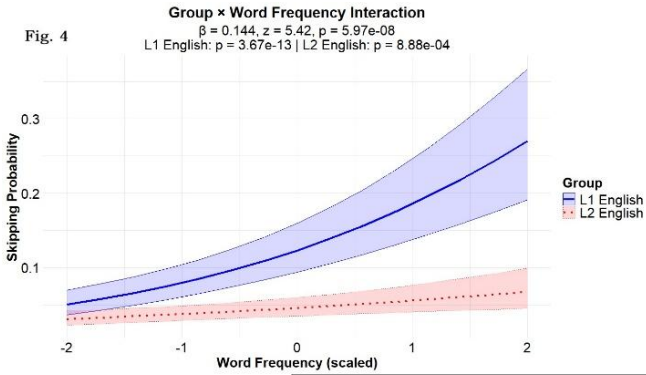
Consistent with previous findings, both oculomotor and linguistic factors were reliable predictors of first-pass skipping. There were higher skipping rates for shorter words (β=-1.76, p<.001) and at shorter launch site distances (β=-1.22, p<.001). High-frequency words were also skipped more often (β=0.33, p<.001), as were words with lower surprisal values (β=-0.22, p<.001). There were also robust group differences related to these effects. L1 readers skipped words more often than L2 readers overall (Fig. 1; β=0.55, p<.001). For the oculomotor factors, the word length effect was larger for L2 readers than for L1 readers (Fig. 2; group × length: β=0.21, p<.001), whereas the effect of launch site distance was larger for L1 readers (Fig. 3; group × launch site: β=-0.13, p<.01). For the linguistic factors, higher word frequency was associated with a larger increase in skipping probability for L1 readers than for L2 readers (Fig. 4; group × frequency: β=0.14, p<.001). Notably, however, there was no reliable interaction of group with GPT surprisal, indicating a relatively consistent influence of predictability on skipping for both L1 and L2 readers (Fig. 5). Importantly also, launch site distance modulated the influence of all other continuous predictors: word length (β=0.12, p<.001), word frequency (β=-0.19, p<.001), and GPT surprisal (β=0.05, p<.01). And the extent of this modulating influence was different for L1 and L2 readers. For example, as illustrated by the heatmaps below (Fig. 6), while there was a larger effect of word length at shorter launch site distances for both L1 and L2 readers, this interaction was particularly strong for the L2 group (group × launch site × word length: β=-0.05, p<.001). A similar pattern of results was found for word frequency. Again, although there was a larger frequency effect at shorter launch site distances for both L1 and L2 readers, this interaction was particularly strong for the L2 group (group × launch site × word frequency: β=0.08, p<.001).

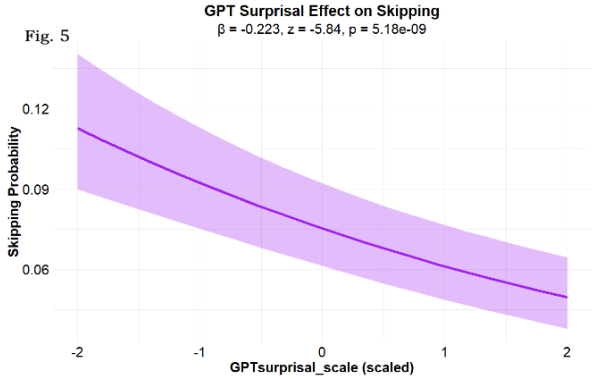
Together, these findings support a model of first-pass word skipping in which low-level oculomotor and higher-level linguistic factors jointly determine skipping behavior in both L1 and L2 reading. However, the evidence also indicates that L1 and L2 readers weigh these factors differently, with a larger influence of word frequency in L1 readers and a larger influence of word length in L2 readers. The findings also underscore the crucial moderating influence of parafoveal preview quality on other factors that influence skipping, particularly for L2 readers. The presentation will discuss the complete set of findings as well as their implications for research into L2 reading and L2 processing more generally.

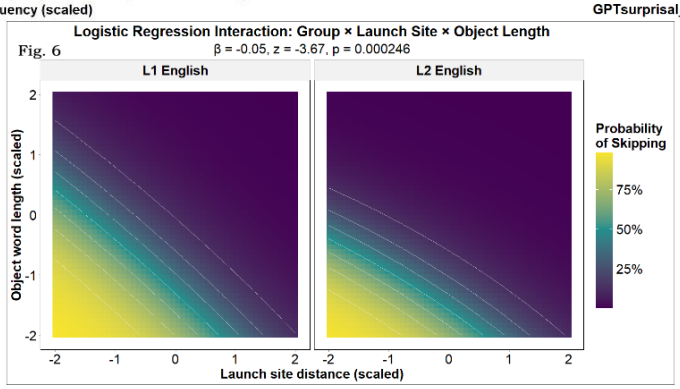












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